Estimating in Building Construction

Chapter 2 Contracts, Bonds, and Insurance

Contract System

- Single contract
  - Owner contracts with prime contractor
  - Prime contractor hires subcontractors and material suppliers
  - Subcontractors and suppliers are responsible towards the prime contractor
  - Prime contractor is responsible directly to the owner.

Contract System

- Separate contracts
  - Owner contracts separately for:
    - General construction
    - Plumbing
    - Heating
    - Electrical
    - Sewage disposal
    - Specialties (Elevators, Lifts, Air conditioning)
    - Etc.

Types of Agreements

- Lump sum (stipulated sum, fixed price)
  - Fixed price or set price through competitive bidding or negotiations.
  - May include change orders which entitled to additional monies for actual work and for overhead, as well as additional time.

Types of Agreements

- Unit-price agreement
  - A price is given for each unit of work
  - Lower bidder is determined based on the owner-provided quantities.
  - Paid based on actual quantity of work
  - Neither the owner nor the contractor will know the exact cost of the project until its completion.

Types of Agreements

- Cost-plus-fee
  - Paid for actual construction cost
  - Plus a fee:
    - Percentage fee
    - Fixed fee
    - Fixed fee with a guaranteed maximum cost (g-max). They share any savings.
    - Sliding scale fee: If the cost of the project increases, the percent fee of the contractor decreases.

Types of Agreements

- Cost-plus-fee
  - Paid for actual construction cost
  - Plus a fee:
    - Fixed fee with a bonus and penalty
    - A target cost estimate is set up; and if the cost is less than the target amount, the contractor receives a bonus in the form of a percentage of the savings. If the cost goes over the target figure, there is a penalty (reduction of percentage).

Agreement Provisions

- Scope of work
- Time of completion
  - Start time
  - Completion time
    - Calendar days
    - Date
  - Liquidated damages

Agreement Provisions

- Contract sum
- Progress payments
  - Due date
- Retained percentage

Agreement Provisions

- Schedule of values
  - Price for specific items
  - Contractor sometimes overvalue the initial items referred to as front-end loading
- Work in place and stored materials
  - How is work completed calculated?
  - Does the owner pay for stored materials?

Agreement Provisions

- Acceptance and final payment
  - Final inspection
  - Certification of completion
  - Acceptance of the work
  - Retainage can be partly released

Bonds

- Surety
  - Guarantees payment on another party's obligations
- Bonding limits based upon
  - Financial reports
  - Experience
  - Work in progress

Bonds

- Bid
  - Submitted with bid
  - Guarantees the employer that the contractor will enter into the contract and provide all other specified bonds.

Bonds

- Performance bonds
  - Guarantees that the contractor will perform all the work in accordance with the contract documents
  - Includes warranty period

Bonds

- Labor and material bond (payment bond)
  - Guarentees contractor will pay for:
    - Labor
    - Materials

Bonds

- Subcontractor bonds
  - Performance, labor and materials bond (payment bond) provided by subcontractor to general contractor
  - Protect prime contractor against financial loss and litigation due to default by a subcontractor
  - Reduces general contractor's risk
  - Preserves general contractor's bonding capacity

Bonds

- License or permit bond
  - State law or municipal ordinance requires a contractor's license or permit.
  - Guarantees compliance with statutes and ordinances

Insurance

- Contractor must have insurance for protection of the assets of their business.

- There must not be gaps in the insurance coverage that might cause the contractor serious financial loss.

- Insurance is not same as bond. With an insurance policy, responsibility of loss is with the insurance company. In contrast, with a bond, the bonding companies will fulfill the obligations of the bond and then turn to the contractors to reimburse the money that they expended on their behalf.

Insurance

- Workers' compensation insurance
  - Provides benefits to workers or their families if they are injured or killed on the job
  - Rate charged for this insurance are based upon:
    - Location of work (state)
    - Type of work
    - Company's claim history

Insurance

- Workers' compensation insurance
  - Worker's compensation benefits include:
    - Lost wages
    - Economic loss
    - Medical expenses
    - Benefits to dependents if someone is killed

Insurance

- Builder's risk fire insurance
  - Covers loss due to fire
  - May be extended to:
    - Other weather related damage
    - Explosions
    - Riots and civil disorder
    - Vandalism and malicious mischief
    - Theft

- **Read Chapter 17 Construction Insurance - Book by Jimmie Hinze – “Construction Contracts”**

Thank You